



The Union-Closed Sets Conjecture

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Overview

- Statement of the Conjecture
- The First Constant Lower Bound
- Entropy and Some Theorems
- Recent Improvements on Bound

What is Union-Closed?

Before we state the union-closed sets conjecture, we shall first go over the definition of union-closed.

Definition (Union-Closed)

A family $\mathcal{F}$ of sets is *union-closed* if for every two member-sets $A, B \in \mathcal{F}$, their union $A \cup B$ is contained in $\mathcal{F}$.

Example

An example of a union-closed family of sets is

$$\mathcal{F} = \{\{1, 2\}, \{2, 3\}, \{1, 2, 3\}, \{1\}, \emptyset\}.$$

The Union-Closed Sets Conjecture

With the definition of union-closed in mind, we will go over the simple, yet extremely challenging, conjecture, posed in 1979.

Statement

Any finite union-closed family of sets $\mathcal{A} \neq \{\emptyset\}$ has an element that is contained in at least 50% of the member-sets.

Example

The union-closed family of sets $\mathcal{F} = \{\{1,2\}, \{2,3\}, \{1,2,3\}, \{1\}, \emptyset\}$ satisfies the union-closed sets conjecture because the element 1 is contained in 3 of the 5 sets, and $\frac{3}{5} = 60\% \geq 50\%$.

The First Constant Lower Bound

Statement

Any finite union-closed family of sets $\mathcal{A} \neq \{\emptyset\}$ has an element that is contained in at least 50% of the member-sets.

Even after over 40 years, we have yet to prove the statement completely (i.e., achieve the bound of 50%). However, all open problems have gradual progress. The first constant lower bound to be discovered was 1%.

What is Entropy?

In order to prove the 1% constant lower bound, the concept of entropy was used. You might know entropy as a scientific concept that is related to randomness and disorder. However, we shall introduce the information theory definition of entropy.

Definition

Given a random variable X with possible values in a set A and a probability distribution p , define the *entropy*, or $H(X)$, by

$$H(X) := - \sum_{x \in A} p(x) \log p(x) = \mathbb{E}[-\log p(X)].$$

Two Theorems with Entropy

Here are two theorems that constitute the 1% bound and make use of entropy.

Theorem 1 (Justin Gilmer)

Let A and B denote independent samples from a distribution over subsets of $\{1, \dots, n\}$. Assume that for all $i \in \{1, \dots, n\}$, $\Pr[i \in A] \leq 0.01$. Then

$$H(A \cup B) \geq 1.26H(A).$$

Theorem 2 (Justin Gilmer)

Let $\mathcal{F} \subseteq \mathcal{P}(\{1, \dots, n\})$ be a union-closed family, $\mathcal{F} \neq \{\emptyset\}$. Then there exists $i \in \{1, \dots, n\}$ that is contained in at least 1% of the sets in $\mathcal{F}$.

Theorem 1 implies Theorem 2

Theorem 1 says that for any $i \in \{1, \dots, n\}$, if the probability that i is in A is at most 1%, then $H(A \cup B) \geq 1.26H(A)$. We also know that $H(A) > 0$, so Theorem 1 implies that $H(A \cup B) > H(A)$. Note that $A \cup B$ is also a distribution over $\mathcal{F}$. Also, the entropy of a distribution is maximized when it is uniform. Intuitively, we see that $A \cup B$ is “further away” from uniform than A , so $H(A \cup B) \leq H(A)$. This contradicts Theorem 1, so Theorem 2 follows.

Recent Improvements on Bound

In 2022, the bound was improved to $\frac{3-\sqrt{5}}{2} \approx 38\%$. This bound was given in three different publications (Ryan Alweiss, Brice Huang, Mark Sellke; Zachary Chase and Shachar Lovett; Will Sawin). However, the 38% is tight for this method, and new techniques will be needed to reach 50%.

Conclusion

Sometimes, deceptively simple problems like the Union Closed Sets Conjecture can prove to be some of math's most challenging. However, new techniques have led to recent progress on this problem.

References



Justin Gilmer.

A constant lower bound for the union-closed sets conjecture.

2022.